Substituting $f = -\frac{1}{2}$ in (6), $g = -\frac{5}{2}$.

Substituting $f = -\frac{1}{2}$ and $g = -\frac{5}{2}$ in (2), $c = 4$.

Therefore, the required equation of the circle is

$$x^2 + y^2 + 2\left(-\frac{5}{2}\right)x + 2\left(-\frac{1}{2}\right)y + 4 = 0$$

$$\Rightarrow x^2 + y^2 - 5x - y + 4 = 0.$$

Note

Three points on a circle determine the equation to the circle uniquely. Conversely three equidistant points from a centre point forms a circle.

5.2.2 Equations of tangent and normal at a point P on a given circle

Tangent of a circle is a line which touches the circle at only one point and normal is a line perpendicular to the tangent and passing through the point of contact.

Let $P(x_1, y_1)$ and $Q(x_2, y_2)$ be two points on the circle $x^2 + y^2 + 2gx + 2fy + c = 0$.

Therefore,

$$x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c = 0 \quad \text{... (1)}$$

and $x_2^2 + y_2^2 + 2gx_2 + 2fy_2 + c = 0$,,, (2)

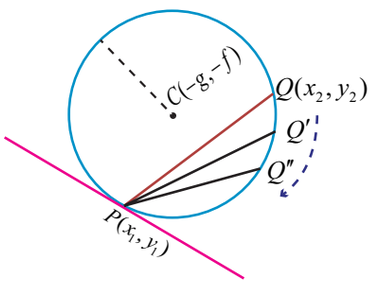


Fig.5.13

(2) − (1) gives

$$x_2^2 - x_1^2 + y_2^2 - y_1^2 + 2g(x_2 - x_1) + 2f(y_2 - y_1) = 0$$

$$(x_2 - x_1)(x_2 + x_1 + 2g) + (y_2 - y_1)(y_2 + y_1 + 2f) = 0$$

$$\frac{x_2 + x_1 + 2g}{y_2 + y_1 + 2f} = -\frac{(y_2 - y_1)}{(x_2 - x_1)}$$

Therefore, slope of $PQ = -\frac{(x_1 + x_2 + 2g)}{(y_1 + y_2 + 2f)}$.

When $Q \to P$, the chord PQ becomes tangent at P

Slope of tangent is $-\frac{(2x_1 + 2g)}{(2y_1 + 2f)} = -\frac{(x_1 + g)}{(y_1 + f)}$.

Hence, the equation of tangent is $y - y_1 = -\frac{(x_1 + g)}{(y_1 + f)}(x - x_1)$. Simplifying,

$$yy_1 + fy - y_1^2 - fy_1 + xx_1 - x_1^2 + gx - gx_1 = 0$$

$$xx_1 + yy_1 + gx + fy - (x_1^2 + y_1^2 + gx_1 + fy_1) = 0 \quad \text{...(1)}$$

Since (x_1, y_1) is a point on the circle, we have $x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c = 0$

Therefore, $-(x_1^2 + y_1^2 + gx_1 + fy_1) = gx_1 + fy_1 + c$. ...(2)

Hence, substituting (2) in (1), we get the equation of tangent at (x_1, y_1) as

$$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0.$$

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